

Zion Tech Hub Capstone Project

Diabetes Risk Prediction Using Machine Learning (Cohort 9.0)

Project Scenario

You are a data scientist working with a healthcare clinic.

The clinic wants to predict whether a patient is likely to have diabetes based on medical data.

Your task is to analyze the dataset, build a prediction model, and explain your findings in simple terms.

Dataset

You are provided with a dataset containing patient health information.

The column 'Outcome' (or 'target') represents:

0 → No diabetes

1 → Has diabetes

Important: The dataset contains missing values (NaN). Handling them properly is part of your task.

Your Tasks

1. Understand the dataset (load, inspect, check types)
2. Identify missing values
3. Handle missing values (explain your method)
4. Perform EDA (3–5 visualizations with insights)
5. Build a model (e.g., Logistic Regression)
6. Evaluate the model using:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix
7. Explain results in simple terms

8. Give final recommendation for healthcare use

Written Report (Google Docs)

Keep it simple (2–3 pages max):

1. Introduction
2. Data Cleaning (what you did and why)
3. Analysis (insights)
4. Model & Evaluation
5. Conclusion

Submission Instructions

Submit on Google Classroom:

1. Google Colab Notebook
2. Google Docs Report

Timeline

Duration: 3 weeks

Hints

- Start simple
- Focus on understanding, not perfection
- Rewatch class recordings if needed
- Do not ignore missing values
- There is no single correct answer — explain your decisions

What the Instructor is Looking For

- Understanding of the problem
- Reasonable decision making
- Clear explanation of work
- Ability to communicate findings